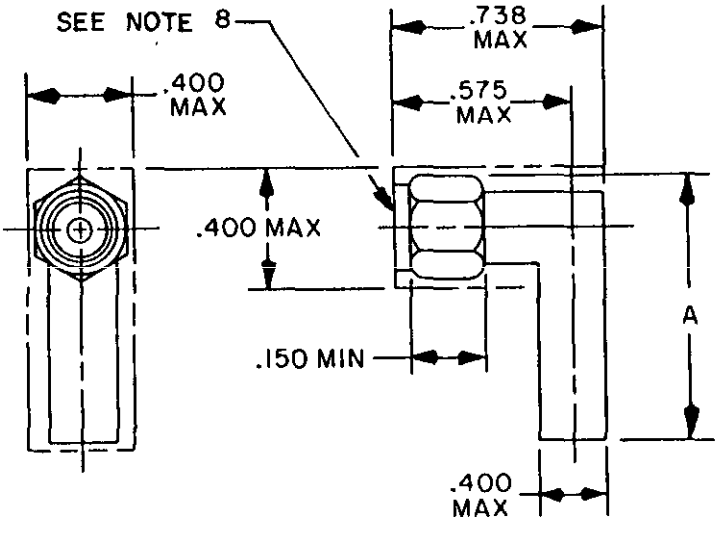
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUGS, ELECTRICAL, COAXIAL, RADIO FREQUENCY,

(SERIES SMA (CABLED) - PLUG, PIN CONTACT, RIGHT ANGLE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



Inches	mm
.150	3.81
.400	10.16
.575	14.60
.738	18.75

MARKING IMPLEMENTATION DATE, CATEGORY B, SEE TABLE VII

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. For dimension A, see table I and V.
4. Dimension .400 (10.16 mm) is the largest overall diameter of the connector.
5. Width across flats are to accomodate wrench, nominal size of .3125 (7.938 mm) minimum in accordance with FED-STD-H28, appendix 10, wrench openings.
6. Dimension A defines the maximum length of the connector when assembled to the appropriate cable.
7. All undimensioned pictorial configurations are for reference purposes only.
8. Series SMA, pin contact interface in accordance with MIL-STD-348.

FIGURE 1. General configuration.

AMSC N/A
FSC 5935

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TABLE I. Dash numbers, cross reference and dimensions.

Dash number	Applicable cable #	Dimensions	Inches-millimeters maximum 1/8/6/
Category A - Field serviceable (no special tools required) 3/ 9/			
3006 3106 7/ 4006 4106 7/	M17/93-RG178 M17/169-00001ø	A	1.125(28.58)
3007 3107 7/ 4007 4107 7/	M17/119-RG174 M17/173-00001ø M17/113-RG316 M17/172-00001ø		
3008 3108 7/ 4008 4108 7/	M17/54-RG122* M17/157-00001ø		
3009 3109 7/ 4009 4109 7/	M17/28-RG058 M17/60-RG142@ M17/84-RG223* M17/155-00001ø M17/158-00001ø M17/167-00001ø		
3010 3110 7/ 4010 4110 7/	M17/111-RG303*@ M17/170-00001ø		
3030 3130 7/ 4030 4130 7/	M17/152-00001		
Category C - Field replaceable (MIL-C-22520 crimp tool) 2/3/			
3025 3125 7/ 4025 4125 7/	M17/93-RG178∞ M17/169-00001∞ø	A	1.375(34.93)
3026 3126 7/ 4026 4126 7/	M17/119-RG174Σ M17/173-00001Σø M17/113-RG316Σ M17/172-00001Σø		
3027 3127 7/ 4027 4127 7/	M17/54-RG122&* M17/157-00001&ø		
3028 3128 7/ 4028 4128 7/	M17/60-RG142@~ M17/158-00001~ø M17/167-00001~ø M17/84-RG223*~		
3029 3129 7/ 4029 4129 7/	M17/155-00001~ø M17/28-RG058*~ M17/111-RG303~ M17/170-00001~ø		

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TABLE I. Dash numbers, cross reference and dimensions - Continued.

Dash number†	Applicable cable#	Dimensions	Inches-millimeters maximum 1/8/6/
Category D - Field replaceable - defined piece parts 2/4/5/3/			
3502	M17/60-RG142*@	A	1.375(34.93)
3602 7/	M17/158-00001ø		
4502	M17/128-RG400		
4602 7/	M17/175-00001ø		

1/ Millimeters are in parentheses.

2/ These connectors are assembled, using the applicable crimp tool, to the specified cables stripped as shown on figure 3.

3/ These connectors have captivated center contacts.

4/ Complete connector assembly shall consist of a body, center contact, ferrule, and assembly instructions.

5/ Not for use in army equipment.

6/ Coupling nuts shall be corrosion resistant steel with a passivated finish per MIL-F-14072.
(applies only to "-3XXX" SERIES CONNECTORS)

7/ No safety wire holes.

8/ For logistics purposes, only connectors with safety wire holes will be stocked.

9/ All corrosion resistant steel bodied connectors which are designed to be assembled to the outer conductor using solder shall be gold plated per MIL-G-45204, Type II, Class 1.

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in note @.

@ Cable to be used for the +200°C temperature cycling tests.

† Connectors mate with connectors of the same material; i.e., M39012/56-3001 mates with M39012/57-3001, and M39012/56-4001 mates with M39012/57-4001.

∞ Preferred die M22520/5-33 closure B, alternate die M22520/5-03 closure B.

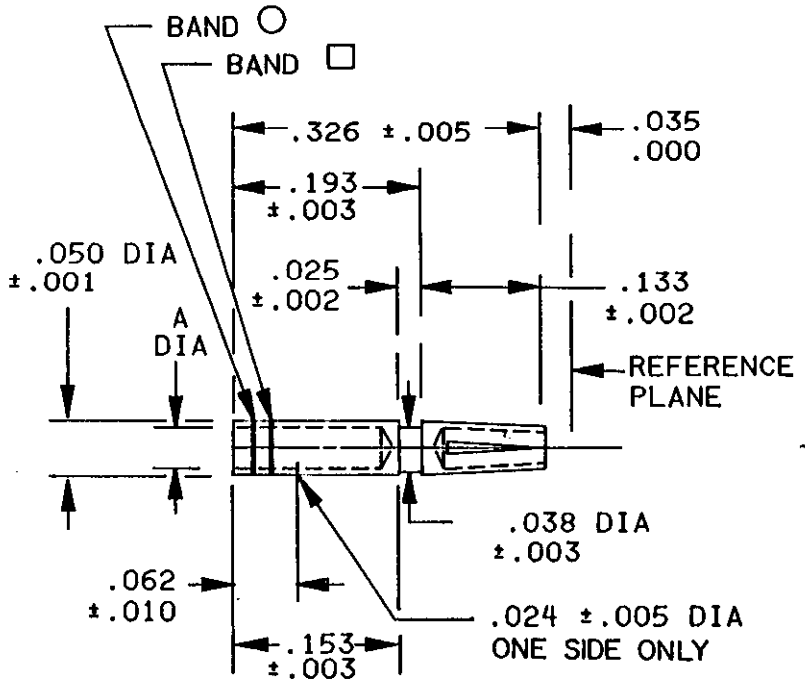
Σ Preferred die M22520/5-35 closure B, alternate die M22520/5-03 closure A.

& Preferred die M22520/5-41 closure B, alternate die M22520/5-05 closure B, or -09 closure A.

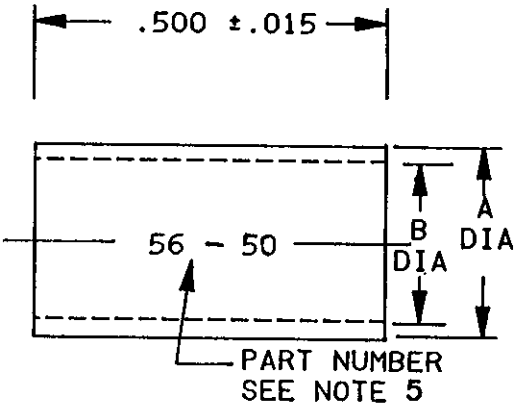
~ Preferred die M22520-5/19 closure B, alternate die M22520/5-05 closure A or -11, -57, closure A.

ø Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

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CENTER CONTACT



CRIMP FERRULE

Dash no.	Contact no. 2/	A ±.001	Basic crimp tool 1/	Crimp die or positioner	Crimp tensile min	Color band □	Color band ○
3502 4502	56-10	.041	M22520/1-01	Solder or M22520/1-15	4 lbs	Red	Maroon

Dash no.	Ferrule no. 2/	A ±.001	B ±.001	Basic crimp tool 1/	Crimp die or positioner
3502 4502	56-50	.250	.220	M22520/5-01	M22520/5- 05, 11, 57 Closure A or 19 Closure B

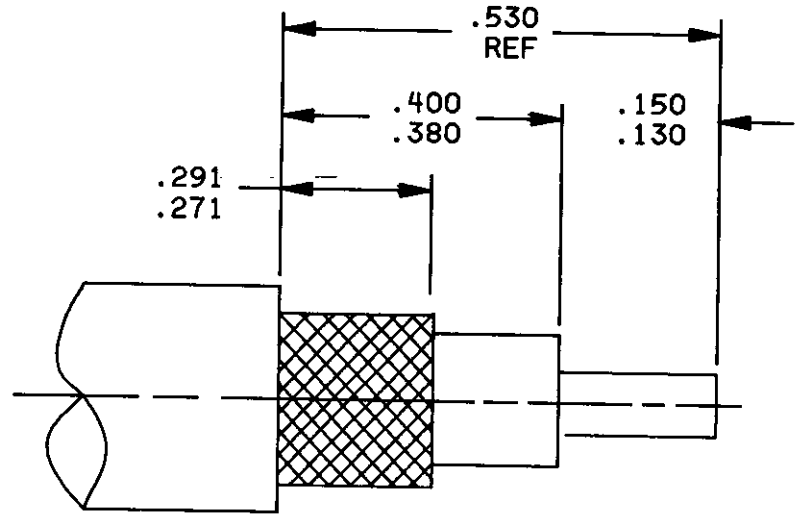
Inches	mm	Inches	mm
.001	0.03	.041	1.27
.002	0.05	.050	1.57
.003	0.08	.062	2.54
.005	0.13	.153	3.89
.010	0.25	.193	4.90
.015	0.38	.220	5.59
.024	0.61	.250	6.35
.025	0.64	.326	8.28
.035	0.89	.500	12.70
.038	0.97		

1/ Class 2 tool may be used by OEM (see MIL-C-22520).
2/ Contact numbers and ferrule numbers are for identification only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Solder or crimp tensile test shall be in accordance with MIL-C-39029.
4. Copyright notice: All information disclosed in these specification sheets which is or may be copyrighted is reproduced herein with the express permission of the copyright owner.
5. Contact numbers and ferrule numbers are for identification only.

FIGURE 2. Contact and ferrule dimensions for category D only.

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Inches	mm
.130	3.30
.150	3.81
.271	6.88
.291	7.39
.380	9.65
.400	10.16
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Cable stripping dimensions for field replaceable connectors.

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 12,400 MHz.

Voltage rating: The voltage rating shall be in accordance with table II.

TABLE II. Voltage rating.

Cables	Voltage max (at sea level)	Voltage max (70,000 ft)
	V rms	V rms
M17/93-RG178, M17/169-00001	170	45
M17/54-RG122, M17/157-00001 M17/119-RG174, M17/173-00001 M17/94-RG179, M17/113-RG316, M17/172-00001 M17/152-00001	250	65
M17/28-RG058, M17/155-00001 M17/60-RG142, M17/158-00001 M17/84-RG223, M17/167-00001 M17/111-RG303, M17/170-00001 M17/128-RG400, M17/175-00001	335	85

Temperature rating: -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 2 inch-pounds, maximum.

Coupling proof torque: 15 inch-pounds, minimum.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

Mating characteristics: Reference MIL-STD-348 for dimensions.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302 of MIL-STD-202, test condition B.

5,000 megohms, minimum.

Center contact retention: 6 pounds, minimum axial force. Applicable to captivated-center-contact connectors only.

Radial torque: Not applicable.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

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Voltage standing wave ratio (VSWR):

Cables	VSWR
M17/93-RG178	1.20+0.03 (F) GHz
M17/54-RG122 M17/119-RG174 M17/94-RG179 M17/113-RG316, M17/152-00001	1.15+0.03 (F) GHz
M17/28-RG058 M17/60-RG142 M17/84-RG223 M17/111-RG303 M17/128-RG400	1.15+0.02 (F) GHz

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.025+.002 F (F in GHz).

Item 16: VSWR shall be less than 1.025+.002 F (F in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.080+.005 F (F in GHz).

Group B inspection: Use step 5, long cable method.

Qualification and group C inspection: Use step 5, long cable method.

Connector durability:

Insertion and withdrawal force: 500 cycles, minimum at 12 cycles per minute, maximum.

The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum.

	Initial	After environment
Center contact:	4.0	6.0
Outer contact:	2.0	Not applicable
Braid to body:	0.5 1/	Not applicable

Dielectric withstanding voltage at sea level:

Method 301 of MIL-STD-202.

Cables	Vrms
M17/93-RG178, M17/169-00001	500
M17/54-RG122, M17/157-00001 M17/119-RG174, M17/173-00001 M17/94-RG179, M17/152-00001 M17/113-RG316, M17/172-00001	750
M17/28-RG058, M17/155-00001 M17/60-RG142, M17/158-00001 M17/84-RG223, M17/167-00001 M17/111-RG303, M17/170-00001 M17/128-RG400, M17/175-00001	1,000

1/ Five milliohms are permissible on passivated steel bodied connectors.

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Vibration, high frequency: Method 204 of MIL-STD-202, test condition D. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables (see tables I and V).

Moisture resistance:

Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Altitude: 70,000 feet.

Cables	Volts (min)
M17/93-RG178, M17/169-00001	125
M17/54-RG122, M17/157-00001 M17/119-RG174, M17/173-00001 M17/94-RG179, M17/152-00001 M17/113-RG316, M17/172-00001	190
M17/28-RG058, M17/155-00001 M17/60-RG142, M17/158-00001 M17/84-RG223, M17/167-00001 M17/111-RG303, M17/170-00001 M17/128-RG400, M17/175-00001	250

RF high potential withstanding voltage:

Frequency: 5 to 7.5 MHz

Leakage current: Not applicable.

Cables	Volts (min)
M17/93-RG178, M17/169-00001	335
M17/54-RG122, M17/157-00001 M17/119-RG174, M17/173-00001 M17/94-RG179, M17/152-00001 M17/113-RG316, M17/172-00001	500
M17/28-RG058, M17/155-00001 M17/60-RG142, M17/158-00001 M17/84-RG223, M17/167-00001 M17/111-RG303, M17/170-00001 M17/128-RG400, M17/175-00001	670

Cable retention force: The cable retention force shall be in accordance with table III.

TABLE III. Cable retention force.

Cable dielectric outer diameter	Pounds (min)	
	Single braid	Double braid
Inches (max)		
.036	10	N/A
.067	20	N/A
.110	30	N/A
.122	40	45

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Coupling mechanism retention force: 60 pounds, minimum.

RF leakage: -60 dB, minimum tested at a frequency between 2 and 3 GHz.

This requirement may be met by meeting the RF leakage requirement on the MIL-C-39016/55 connector which has the same mating interface design.

RF insertion loss: dB max = .15 x $\sqrt{\text{freq GHz}}$. Test frequency at 6.0 GHz.

Part number: M39012/56 (dash number from table I or "B" number from table V).

TABLE IV. Group qualification and retention testing.

Group	Submission and qualification of any of the following connectors 1/ 2/	Qualifies the following connectors
I	M39012/56-†009	M39012/56-†006 M39012/56-†007 M39012/56-†008 M39012/56-†009 M39012/56-†010 M39012/16-†030
II	M39012/56B†015	M39012/56B†011 M39012/56B†012 M39012/56B†013 M39012/56B†014 M39012/56B†015 M39012/56B†016 M39012/56B†017
III	M39012/56B†02?	M39012/56B†018 M39012/56B†019 M39012/56B†020 M39012/56B†021 M39012/56B†022 M39012/56B†023 M39012/56B†024
IV	M39012/56-†028	M39012/56-†025 M39012/56-†026 M39012/56-†027 M39012/56-†028 M39012/56-†029
V	M39012/56-†502	M39012/56-†502

1/ Individual connectors other than listed are self qualifying only.
2/ Qualification of connectors qualifies connectors of the same material only.
3/ Connectors qualified with safety wire holes automatically qualifies connectors without safety wire holes.

† Denotes material.

NOTES:

1. For qualification retention, where more than one part is listed in a group in this column, data may be supplied on any of those parts in order to retain qualification for those parts in the corresponding right hand column. The part does not necessarily have to be the part initially qualified.

2. If a connector manufacturer produces a connector which meets all the requirements for two or more connector part numbers (within the same series), the manufacturer may receive qualification approval for two or more connector part numbers by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number. For group qualification, the connectors must be of similar design.

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TABLE V. Category B - Nonfield replaceable (special tools may be required).

Not for Air Force or Navy use. For OEM use only.

M39012/56B	Applicable cable #	Dimensions	Inches-millimeters maximum 1/8/9/10/
3011 3/	M17/93-RG178		
3111 4/ 3/	M17/169-00001ø		
4011 3/			
4111 4/ 3/			
3012 3/	M17/119-RG174		
3112 4/ 3/	M17/173-00001ø		
4012 3/	M17/113-RG316		
4112 4/ 3/	M17/172-00001ø	A	1.250(31.75)
3013 3/	M17/54-RG122*		
3113 4/ 3/	M17/157-00001ø		
4013 3/			
4113 4/ 3/			
3014 3/	M17/28-RG058*		
3114 4/ 3/	M17/155-00001ø		
4014 3/			
4114 4/ 3/			
3015 3/	M17/60-RG142*@		
3115 4/ 3/	M17/158-00001ø		
4015 3/			
4115 4/ 3/			
3016 3/	M17/84-RG223*		
3116 4/ 3/	M17/167-00001ø		
4016 3/			
4116 4/ 3/			
3017 3/	M17/111-RG303*@		
3117 4/ 3/	M17/170-00001ø		
4017 3/			
4117 4/ 3/			
3018 2/	M17/93-RG178		
3118 4/ 2/	M17/169-00001ø		
4018 2/			
4118 4/ 2/			
3019 2/	M17/119-RG174		
3119 4/ 2/	M17/173-00001ø		
4019 2/	M17/113-RG316		
4119 4/ 3/	M17/172-00001ø		
3020 2/	M17/54-RG122*		
3120 4/ 2/	M17/157-00001ø		
4020 2/			
4120 4/ 2/			
3021 2/	M17/28-RG058*	A	1.375(34.93)
3121 4/ 2/	M17/155-00001ø		
4021 2/			
4121 4/ 2/			
3022 2/	M17/60-RG142*@		
3122 4/ 2/	M17/158-00001ø		
4022 2/			
4122 4/ 2/			

See footnotes at end of table.

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TABLE V. Category B - Nonfield replaceable (special tools may be required).

Not for Air Force or Navy use. For OEM use only.

M39012/56B	Applicable cable #	Dimensions	Inches-millimeters maximum 1/5/6/7/
3023 2/ 3123 4/ 2/ 4023 2/ 4123 4/ 2/	M17/84-RG223* M17/167-00001ø	A	1.375(34.93)
3024 2/ 3124 4/ 2/ 4024 2/ 4124 4/ 2/	M17/111-RG303 M17/170-00001ø	A	1.375(34.93)

1/ Millimeters are in parentheses.

2/ These connectors have captivated center contacts.

3/ Inactive for new design.

4/ No safety wire holes.

5/ For logistics purposes, only connectors with safety wire holes will be stocked.

6/ All corrosion resistant steel bodied connectors which are designed to be assembled to the outer conductor using solder shall be gold plated per MIL-G-45204, Type II, Class 1.

7/ Coupling nuts shall be corrosion resistant steel with a passivated finish per MIL-F-14072.
(applies only to "-3XXX" SERIES CONNECTORS).

The latest version of each cable shall be applicable.

* Cable to be used when performing tests requiring cable except as in note @.

@ Cable to be used for the +200°C temperature cycling tests.

Connectors mate with connectors of the same material; i.e., M39012/56-3001 mates with M39012/57-3001, and M39012/56-4001 mates with M39012/57-4001.

ø Caution is directed to the application of this cable above 400 MHz. Attenuation is tested only at 400 MHz. SRL and power handling capabilities are not stipulated herein.

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TABLE VI. Maintenance replacements for category B.

Category B number*	Category C dash number	Category A dash number	Category D dash number
B†011	†025	†001,†006	---
B†012	†026	†002,†007	---
B†013	†027	†003,†008	---
B†014	†029	†004,†009	---
B†015	†028	†004,†009	†502
B†016	†028	†004,†009	---
B†017	†029	†005,†010	---
B†018	†025	†006	---
B†019	†026	†007	---
B†020	†027	†008	---
B†021	†029	†009	---
B†022	†028	†009	†502
B†023	†028	†009	---
B†024	†029	†010	---

*Category B connectors are for original installation only. They will not be stocked or procured by the Government.

† The material of the item shall be the same material as the item being replaced. Example: 55-3011 (corrosion resistant steel) replaces 55-3025.

TABLE VII. Cross-reference of part numbers.

Part number M39012/56B 1/	Superseded part number M39012/56-
†011	† 011
†111	† 111
†012	† 012
†112	† 112
†013	† 013
†113	† 113
†014	† 014
†114	† 114
†015	† 015
†115	† 115
†016	† 016
†116	† 116
†017	† 017
†117	† 117
†018	† 018
†118	† 118
†019	† 019
†119	† 119
†020	† 020
†120	† 120
†021	† 021
†121	† 121
†022	† 022
†122	† 122
†023	† 023
†123	† 123
†024	† 024
†124	† 124

See footnotes at end of table.

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